YouTube Shadowing Studio — User Manual

Welcome to **YouTube Shadowing Studio**, a web-based interactive learning environment designed for English language shadowing, typing practice, and audio recording.

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1. Overview & Core Features

YouTube Shadowing Studio allows language learners to practice listening, speaking, and typing using YouTube content.

- **Split-View Workflow**: Watch videos on the left side while practicing shadowing, typing, or writing script notes on the right panel.
- **Smart Time-Bookmark & Memory**: Custom time timestamps (e.g., `33:00`) are stored locally, so even non-embeddable videos can preserve your study progress.
- **Voice Recorder & Visualizer**: Record your voice while speaking along with the video and listen to immediate audio playback.
- **Real-time Typing Accuracy Evaluator**: Practice sentence dictation with real-time accuracy scoring and color-coded character feedback.
- **Local Persistence**: Saved videos and playback positions are stored directly in your browser (`localStorage`).

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2. Getting Started

1. **Opening the Application**: Open `index.html` in any modern web browser (Google Chrome, Microsoft Edge, Apple Safari, or Mozilla Firefox).
2. **Loading a Video**:
    - Paste a valid YouTube video URL or Video ID (e.g., `https://www.youtube.com/watch?v=dQw4w9WgXcQ` or `dQw4w9WgXcQ`) into the top input bar.
    - Click the **불러오기 (Load)** button or press `Enter`.
    - Alternatively, click **데모 영상 로드 (Load Demo Video)** on the video placeholder screen.

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3. Video Player & Navigation Controls

The main media section includes a embedded YouTube player and customized playback controls:

- **Play / Pause (`Space`)**: Start or pause video playback.
- **Seek -5s / +5s (`Ctrl + Left/Right Arrow`)**: Instantly skip backward or forward by 5 seconds for repetition.
- **Custom Time Jump & Save (`customTimeInput`)**:
    - Type a timestamp in `MM:SS` format (e.g., `33:00`) or in total seconds (e.g., `1980`).
    - Click the red **이동 (Move)** button or press `Enter`.
    - *Note*: Clicking **[이동]** automatically updates and saves the video's start timestamp in your library.
- **A-B Loop Segment (`loopBtn`)**: Toggle a 5-second repeating loop from your current playback point.
- **Playback Speed (`speedSelect`)**: Adjust video speed between `0.5x`, `0.75x`, `1.0x` (default), `1.25x`, and `1.5x`.
- **Progress Bar**: Drag the timeline slider to jump to any point in the video.

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4. Interactive Practice Panel

The right-side panel contains three distinct practice modes accessed via tabs:

Shadowing & Voice Recording

- **Live Audio Visualizer**: Provides visual feedback on microphone input.

- **Start Recording (`recordBtn`)**: Click to begin recording your shadowing practice.
- **Stop Recording (`stopRecordBtn`)**: Stop recording and generate an audio preview file.
- **Audio Playback List**:
  - Listen back to recorded clips directly in the application.
  - Download recordings in compatible audio formats (`.m4a` / `.webm`).
  - Delete unnecessary clips.

Typing Practice & Feedback

- **Target Sentence Display**: View the active sentence to dictate.
- **Typing Box (`typingInput`)**: Type what you hear in English.
- **Real-time Accuracy Badge**: Displays your percentage score (0% – 100%).
- **Diff Preview**: Color-coded feedback:
  - `<span style="color:#10b981; font-weight:bold;">Green</span>`: Correct character.
  - `<span style="color:#f87171; font-weight:bold; decoration:underline;">Red Underline</span>`: Incorrect character.

Script & Notes Management

- **Script Entries**: View timestamped sentences accompanying the lesson.
- **Jump to Timestamp**: Click any timestamp button (e.g., `00:05`) to skip the player to that specific time or open the YouTube video externally.
- **Send to Typing Practice**: Click **"타이핑 연습으로 선택"** to load a script line as your active typing target sentence.
- **Add Custom Sentences**: Click **"+ 문장 추가"** to add custom study sentences with your current video timestamp.

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5. Saved Video Library (Manager)

Located at the bottom of the workspace, the Video Library keeps track of all videos you study.

- **Status Indicators**:
  - **재생 가능 (Playable)**: Videos that can be played inside the embedded frame.
  - **재생 불가 (Unplayable)**: Videos blocked from external embedding by YouTube creators.
- **Saved Start Positions**: Displays the last saved start time (e.g., `시작 위치: 33:00`).
- **Filters**: Sort library cards by `전체 보기 (All)`, `재생 가능 (Playable)`, or `재생 불가 (Unplayable)`.
- **Card Actions**:
  - **학습하기 (Study)**: Reloads the video at its saved start position.

- **YouTube Button**: Opens the video directly on YouTube starting at the saved timestamp.
- **Trash Can**: Permanently deletes the card from your local storage.

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6. Keyboard Shortcuts

Action	Shortcut
Play / Pause Video	`Space`
Rewind 5 Seconds	`Ctrl` + ←
Forward 5 Seconds	`Ctrl` + →
Start / Stop Voice Recording	`Alt` + `R`

Note: Keyboard shortcuts are disabled when typing inside input fields or text areas.

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7. Troubleshooting & FAQs

Q1. Why does a video show "재생 불가" (Unplayable)?

Some YouTube video owners disable "Playback on other websites" (embedding). You can still save custom timestamps in your library and use the YouTube external button to open and watch the video at your saved position.

Q2. Why is my microphone not recording?

1. Ensure your browser has permission to access your microphone.
2. Ensure you are running the application on `localhost` or a secure `https://` origin, as browsers block audio recording over insecure `http://` protocols.

Q3. Where are my saved videos and recordings stored?

Video library metadata and timestamps are stored locally in your browser's `localStorage`. Audio recordings are kept in browser memory during your session and can be downloaded to your device as `.m4a` or `.webm` files.